Gravity – A non-contact force (p263-267)

What is gravity?

1. What is gravity?

2. What happens to the force of gravity when the objects get bigger?

3. Is the force of attraction between you and the Earth small or large?

4. Why is the force of gravity between you and you schoolbag small?

Gravitational fields

1. Why is gravity described as a non-contact force?

2. Draw a diagram to show the Earth's gravitational field.

3. What happens to an object inside the Earth's gravitational field?

4. What happens to the strength of the gravitational field as you get further away from Earth?

5. What happens to planets that lie within the sun's gravitational field?

Comparing mass and weight

What is mass?

1. What is mass?

2. If you travelled to another planet, what would happen to your mass?

3. Name three units, and their symbols, used to measure mass.

What is weight?

1. What is weight?

2. What is weight measured in?

3. What does weight depend on?

4. How does your weight differ on the moon?

Falling

1. For many centuries, people believed that heavy things fell faster than light things. Why did they believe this?

2. What did Galilei Galileo realise?

3. What happens as an object falls?

4. Why does a leaf fall slower than a stone of the same mass?

Terminal Velocity

1. What happens as an object's speed increases?

2. What happens when the air resistance equals the weight force?

3. What is this speed called?

4. Why do skydivers need to use parachutes?

Flight

1. There are four key forces at work on an aircraft. Draw a simple diagram of an aircraft and label the four forces on your diagram.

2. Describe each of the four forces:
   i. Weight

   ii. Lift

   iii. Drag

   iv. Thrust

3. The shape of the wings, called an aerofoil, causes a lift force. Explain the shape of the wing and how this creates lift.